

BENCHAOU Nada

STATE-CERTIFIED ENGINEER IN DIGITAL HEALTH
ARTIFICIAL INTELLIGENCE • DATA SCIENCE • SOFTWARE ENGINEERING



bnadabenchou@gmail.com

+212 6 82 29 11 14

Casablanca, Morocco

GitHub : [BNAD-A](#)

LinkedIn: [Benchaou Nada](#)

About Me

As a Digital Health Engineer, my mission is to shape the future of healthcare through technology. From designing robust data architectures to developing Machine Learning and Deep Learning models, I enjoy solving meaningful technical challenges that make healthcare systems smarter, more efficient, and more reliable.

Driven to transform healthcare, one line of code at a time.

Education

ENGINEERING IN DIGITAL HEALTH	
École Supérieure Mohammed VI d'Ingénieurs en Sciences de la Santé	Casablanca, Morocco 2023-2026
PREPARATORY CLASSES IN BIOMEDICAL ENGINEERING	
École Supérieure Mohammed VI d'Ingénieurs en Sciences de la Santé	Casablanca, Morocco 2021-2023

Experiences

FEBRUARY – AUGUST 2026	FINAL YEAR ENGINEERING INTERNSHIP
FIRETHUNDER, CASABLANCA, MOROCCO SOFTWARE ENGINEERING DIVISION	FIREPACSV2 Designed and developed FirePACSV2, a next-generation PACS platform integrating Artificial Intelligence, Deep Learning, Vision-Language Models (VLM), Retrieval-Augmented Generation (RAG), and DICOM/HL7/FHIR interoperability.
JULY - AUGUST 2025	TECHNICAL INTERNSHIP
CHU - OUJDA, MOROCCO IT AND INFORMATION SYSTEMS DEPARTMENT	Double Regard Development of an AI-based system for retinal imaging, including the integration of attention modules (Double CBAM) to enable high-precision automated detection.
JUILLET 2024	OBSERVATION INTERNSHIP
CHU - OUJDA, MOROCCO BIOMEDICAL DEPARTMENT	- Observed surgeries, performed respirator self-tests, and assisted in equipment maintenance - Visited intensive care, neonatology, and radiology departments, and assisted in the installation and configuration of an MRI system

Projects

- Eklia** – AI ASSISTANT FOR RADIOLOGISTS: BRAIN MRI DETECTION & AUTOMATIC REPORT GENERATION
- Designed an intelligent system combining tumor classification, 3D segmentation, automated medical report generation from multimodal MRIs, and developed a user-friendly web interface for report visualization, validation, and editing.
 - Integrated clinical and imaging data using multimodal fusion, achieving a Dice score > 0.85.

- Rescue Stream** – INTELLIGENT MEDICAL DISPATCHING & BUSINESS INTELLIGENCE SYSTEM
- Designed and developed a decision-support platform integrating a Data Warehouse, Big Data pipelines, and Power BI dashboards to optimize ambulance dispatching and analyze emergency healthcare service performance.

- FineMED - LLM** – ANALYSIS AND RISK MITIGATION OF MEDICAL LARGE LANGUAGE MODELS
- Engineered a platform leveraging a fine-tuned LLM (QLoRA) to analyze and mitigate risks associated with AI-generated medical responses, delivering reliable, safe, and context-aware outputs.

Volunteer Experience

- Volunteer at Association Tadamoun Al Amal: Participated in charity actions during Ramadan
- Organized a humanitarian action with the University Student Union (BDE) in 2020 to help homeless people

Skills

- TECHNICAL**
- Data Science & IA**
Machine Learning, Deep Learning, Data Mining, RAG, LLM
▸ Frameworks: PyTorch, TensorFlow/ Keras, Scikit-learn
- Programming Languages**
Python, C/C++, Matlab
- Databases & Big Data**
SQL, MySQL, MongoDB, Hadoop, Spark, Kafka, Airflow
- Web Development & Deployment**
HTML, CSS, JavaScript, FastAPI
- Tools & Productivity**
GitHub, Office 365, Power BI
- Healthcare Technologies**
DICOM, HL7 v2, FHIR R4, Orthanc
- SOFT SKILLS**
- Analytical thinking and critical reasoning
 - Interdisciplinary collaboration
 - Creativity and innovation mindset
 - Rigor and attention to detail

Langues

- Arabic – Native
- French – Fluent
- English – Upper-Intermediate
- Spanish – Intermediate
- German – Basic

Centres d'intérêts

- Artificial Intelligence
- Robotics
- Digital Health Technologies
- Travel



ONLINE PORTFOLIO —
PROJECTS & ACHIEVEMENTS